

OPERATING INSTRUCTIONS



CapTrak

Product revision 02



Accessories

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About this document

This document describes how to set up the *CapTrak* hardware and give basic information about the *CapTrak* software. Together with the **software help** they form an integral part of *CapTrak* and must be precisely adhered to in order to ensure that *CapTrak* is used as intended and operated correctly to guarantee the concomitant safety of test subjects, users and third parties. Make sure that this document is always available to the users.

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Target group of this document

This document is intended for users in the psychological and neurophysiological research area as well as physicians and medical experts with experience in performing physiological data acquisition.

Structure of this document

This document is divided into the following chapters:

- ▶ [Chapter 1](#) contains the scope of delivery and gives an overview of the hardware.
- ▶ [Chapter 2](#) describes how *CapTrak* is used.
- ▶ [Chapter 3](#) contains information on installing and using the software.
- ▶ [Chapter 4](#) describes the maintenance of the hand-held scanner. This chapter also contains notes on cleaning and disposal of old units.

Revision history

Page	Status	Change
9	modified	manufacturer information updated
9	new	information about EMC testing added
10	new	several new markings with explanations added
24	modified	information about storage updated
33	modified	information about disposal updated
35	new	information about result calculation added

Conventions in this document

Typographical conventions

Bold	indicates items on the user interface (menus, buttons, switches, connectors, options) and is used for emphases in the text
<i>Italic</i>	indicates titles of dialog boxes/tabs, file locations and is used to indicate product names
<u>Underscore</u>	indicates cross-references and web addresses
Monospaced	indicates text or characters to be entered at the keyboard

Symbols



Caution: This symbol indicates that incorrect use of the product(s) may result in a **personal injury** to the test subject, the user and/or a third-party. Failure to observe the information in this document constitutes incorrect use.



Notice: This symbol indicates that the incorrect use of the product(s) may bring about a risk of **damage to property**. Failure to observe the information in this document constitutes incorrect use.



Note or Tip: This symbol draws your attention to important information relating to the current topic and to recommendations on how to use the product(s).



Cross-reference: This symbol indicates a reference to a related chapter, section or document.



New: This symbol indicates changes or new content at this point.

Support and reporting errors

We would ask you to report without delay any error you find in this document, any fault on the products or any malfunction that you observe when using this product. To do so, please contact your local dealer, who will also assist you in general questions about the product.



About CapTrak

For the spatial localization of EEG-electrodes there are already several systems on the market. Although the great majority of these systems deliver reliable data, the disadvantages prevail if you consider time-consuming registrations of each electrode and complicated operation.

That is why Brain Products developed *CapTrak*, a tailor-made localization tool for Brain Products active electrodes. It is quick to apply, accurate, and delivers precise, reproducible results.

CapTrak consists of a light hand-held scanner that integrates two cameras and an intuitive, easy-to-use software. *CapTrak* innovatively employs a triangulation method which allows to detect the positions of the *active electrodes* within a very short time. Determining 160 electrode positions can be done in less than 5 minutes.

Product identification

Product designation	BrainVision CapTrak
Article number	BP-125-1000
Manufacturer	Brain Products GmbH Zeppelinstrasse 7 DE-82205 Gilching Tel.: +49 8105 73384 – 0 Fax: +49 8105 73384 – 505 Website: http://www.brainproducts.com Email: techsup@brainproducts.com

We are registered as a manufacturer of electrical and/or electronic equipment with the stiftung elektro-altgeräte register, Nordostpark 72, 90411 Nürnberg, Germany, under the registration number DE89975135.

CE marking	The Brain Products GmbH confirms the electromagnetic compatibility (EMC) of this product according to the Directive 2014/30/ EU of the European Parliament and of the Council of 26 February 2014 on the harmonisation of the laws of the Member States relating to electromagnetic compatibility.
-------------------	--

The Brain Products GmbH confirms the RoHS compliance of this product according to Directive 2011/65/EU of the European Parliament and the Council of 8 June 2011 on the restriction of the use of certain hazardous substances in electrical and electronic equipment (recast published in the Official Journal of the European Union on 1 July 2011) as well as all its amendments up to and including the Commission delegated directive (EU) 2015/863 of 31 March 2015 (published in the Official Journal of the European Union on 4 June 2015).

EMC tested according to EN 61326-1:2013 (IEC 61326-1:2012)



UKCA marking

Brain Products GmbH confirms the electromagnetic compatibility (EMC) of this product according to the Electromagnetic Compatibility Regulations 2016 of 15 November 2016.

Brain Products GmbH confirms the RoHS compliance of this product according to The Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment Regulations 2012 of 15 November 2016.

Warranty	http://brainproducts.com/contact.php
-----------------	---

Combination with other products

CapTrak is permitted to be combined with the active electrode system, the actiCAP ControlBox and actiCHamp or a LiveAmp adapter for actiCAP.

Beside this general statement about permitted product combinations the user must check, whether all stipulations of each product (e.g. regarding its MR compatibility) are fulfilled for the specific combination and purpose of application (i.e. intended use and correct use). Please observe the operating instructions of the respective products.

Markings on the product

The following labels are affixed to the product:



The Brain Products GmbH confirms the electromagnetic compatibility (EMC) of this product according to the Directive 2014/30/ EU of the European Parliament and of the Council of 26 February 2014 on the harmonisation of the laws of the Member States relating to electromagnetic compatibility.

The Brain Products GmbH confirms the RoHS compliance of this product according to Directive 2011/65/EU of the European Parliament and the Council of 8 June 2011 on the restriction of the use of certain hazardous substances in electrical and electronic equipment (recast published in the Official Journal of the European Union on 1 July 2011) as well as all its amendments up to and including the Commission delegated directive (EU) 2015/863 of 31 March 2015 (published in the Official Journal of the European Union on 4 June 2015).



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Brain Products GmbH confirms the RoHS compliance of this product according to The Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment Regulations 2012 of 15 November 2016.



MR Unsafe: Products with this symbol are not safe for use in an MR environment



MR Conditional: An item with demonstrated safety in the MR environment within defined conditions. At a minimum, address the conditions of the static magnetic field, the switched gradient magnetic field and the radio frequency fields. Additional conditions, including specific configurations of the item, may be required.*



MR Safe: An item that poses no known hazards resulting from exposure to any MR environment. MR Safe items are composed of materials that are electrically non-conductive, non-metallic, and non-magnetic.*



No contact with liquids: Make sure that liquids do not enter the enclosure.



Adhere to the manual.



This symbol confirms compliance with the environmental requirements for electronic products (only applies to China).



This symbol indicates that defective products must not be disposed of with household waste. Dispose of in accordance with national regulations or return the product and its accessories to the manufacturer.



The name and address of the product manufacturer are specified next to this label.

* Definitions from ASTM international standard F2503-20 (IEC 62570:2014)



Safety instructions

Please read the following safety information carefully since it will help prevent personal injury and damage to property. It is assumed you have the required specialist knowledge in handling the product and accessories.

Brain Products will not accept any liability for loss or damage resulting from a failure to follow these operating instructions and, in particular, the safety instructions.

Intended use

CapTrak is intended to localize the positions of actiCAP electrodes on a test subject's head.

CapTrak is not a medical device and may be used in the context of non-medical applications in order to carry out fundamental or applied research on the basis of neurophysiological methodology.

Use of CapTrak for diagnosis, therapy, monitoring of vital physiological processes or other medical purposes is expressly forbidden.

Correct use

CapTrak may be used by personnel in the psychological and neurophysiological research area as well as physicians and medical experts. It must not be used by unqualified persons (e.g. laymen), and persons who cannot read (e.g. due to visual impairment) or understand (e.g. due to a lack of language skills) the operating instructions.

CapTrak is permitted to be used for healthy and sick adults, children and animals.

Provided that all stipulations regarding the correct use are met and that the products are used in accordance with their intended use, CapTrak is permitted to be used in the following environments:

- ▶ research institutes and other non-medical environments (e.g. at home)
- ▶ hospitals, clinics and other medical environments.

CapTrak must not be used in the following environments:

- ▶ MR scanner environment
- ▶ vicinity of explosive gases as may be the case in operating theaters, for example

- ▶ oxygen enriched atmospheres
- ▶ underwater (e.g. sea, swimming pool, bath tub) or in environments in which significant amounts of water could enter components of CapTrak (e.g. under shower, under water-tap).

The user is solely liable for any risks to test subjects associated with the investigation if the product is not used in accordance with the correct use.

General precautions

Handle the product and its accessories with care!

Do not drop the product or allow it to fall and avoid impacts.

Always position the hand-held scanner in such a way that it cannot drop. If the scanner drops, it can lose its calibration. This can result in imprecise measurements. After dropping, check the calibration.

Heat, direct sunlight (UV radiation), moisture, dust, liquids, conductive foreign matter and excessive radiation can shorten the lifetime of the product.

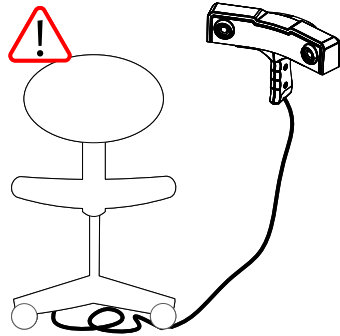


Note

When the hand-held scanner is not in use, store it in the supplied suitcase in a dry place.

**NOTICE****Damage to the hand-held scanner**

- ▶ Do not touch the lenses.
- ▶ Do not allow water to enter the hand-held scanner.
- ▶ Do not pull on the cable.
- ▶ Do not kink or crush cables.





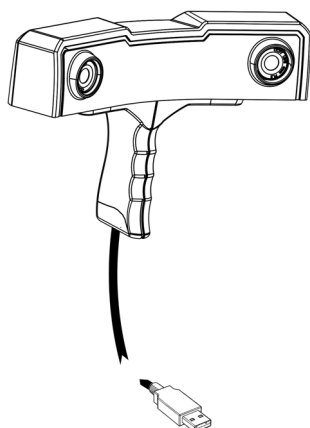
Chapter 1 Getting started

1.1 Scope of delivery

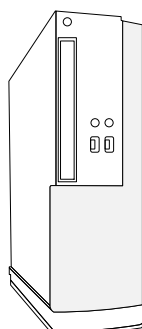
CapTrak is delivered as ready-to-use product including a pre-installed computer and further accessories. The following items are in the scope of delivery:

Hand-held scanner set (BP-125-1001)

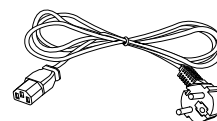
CapTrak hand-held scanner incl. calibration file
BP-225-1001



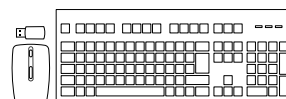
Computer
BP-250-1020



Power cord
(country-specific)
BP-02648xx



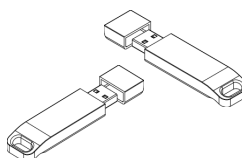
Wireless keyboard and mouse
BP-350-1010



Anti-slip mat (40 x 60 cm), white
BP-225-9020

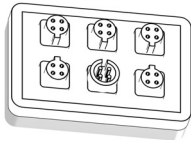


License dongles and Application suite
BP-170-6000 (*CapTrak*)



Landmark set

Ground distributor
BP-245-510



Ground cable
BP-245-1650



Landmarks
BP-225-9011

Self-adhesive rings
FMS-060231



Calibration set

Calibration check board
BP-225-9000



1.2 Functional description

The hand-held scanner of *CapTrak* features two digital cameras, that are arranged in a fixed angle to each other. The scanner has the task to send the images to the associated software.

The core of *CapTrak* is the software: It communicates with active EEG electrodes (via an input equipment) and analyzes the images received from the scanner.

First, the software sends a command to the active EEG electrodes to switch on and off LEDs in the electrodes. The LEDs are switched on in patterns. When the software receives the images from the scanner, it analyzes them for green, red and blue LEDs (that is, light spots). Finally, when the software detects a pattern it switches on the consecutive pattern. At the end of the scanning process all patterns are put together to calculate the precise position of each electrode.

The calculation is based on a reference system built by landmarks. The landmarks are placed on anatomical fiducial points and are also scanned.

The scanning information is then presented on an intuitive user interface in the software.

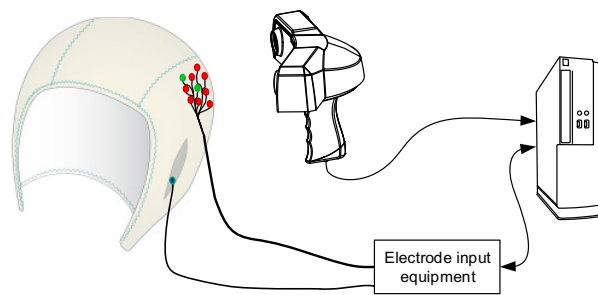


Figure 1-1. Functional principle at a glance



Chapter 2 Using the scanner

2.1 Before the first use

The hand-held scanner was calibrated before delivery. Due to extreme temperatures or shocks on the packing during transport, the hand-held scanner can go out of calibration.

Therefore, check the calibration before the first use. If the calibration check is not successful, contact your local dealer.

Please refer to [Section 4.1](#), for instructions on checking the hand-held scanner.

2.2 Setup with actiCHamp and actiCAP ControlBox

Prepare the following:

- ▶ landmark set (supplied)
- ▶ hand-held scanner (supplied)
- ▶ *actiCHamp* or *actiCAP ControlBox* (incl. USB cable; not supplied)
- ▶ ground electrode and reference electrode¹ (not supplied)
- ▶ electrode cap (incl. splitter boxes; not supplied)
- ▶ USB license dongle for CapTrak (supplied).

1. Only when used with the *ControlBox*.

Do the following:

- 1 Connect the hand-held scanner to the computer (rear USB 3.0 port).
- 2 Plug in the USB license dongle.

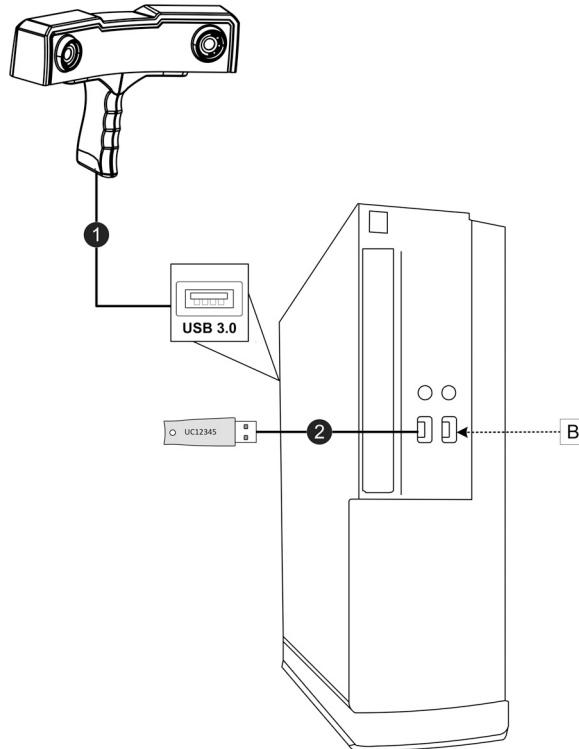


Figure 2-1. Connecting the scanner and USB license dongle

- 3 Connect the ground and three landmarks to the *ground distributor*.

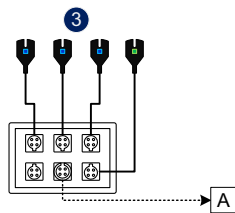


Figure 2-2. Connecting the landmarks and ground electrode

- 4 Connect the *ground distributor* to *actiCHamp* or the *actiCAP ControlBox* (**GND**) with the ground cable (A).
- 5 Connect the flat-ribbon cables of the electrode cap to the *actiCHamp* or the *actiCAP ControlBox*.
- 6 Connect the reference electrode if you are using the *actiCAP ControlBox*.
- 7 Connect *actiCHamp* or the *actiCAP ControlBox* to the computer with the USB cable (B).
- 8 Switch on the computer (and the *actiCAP ControlBox*).

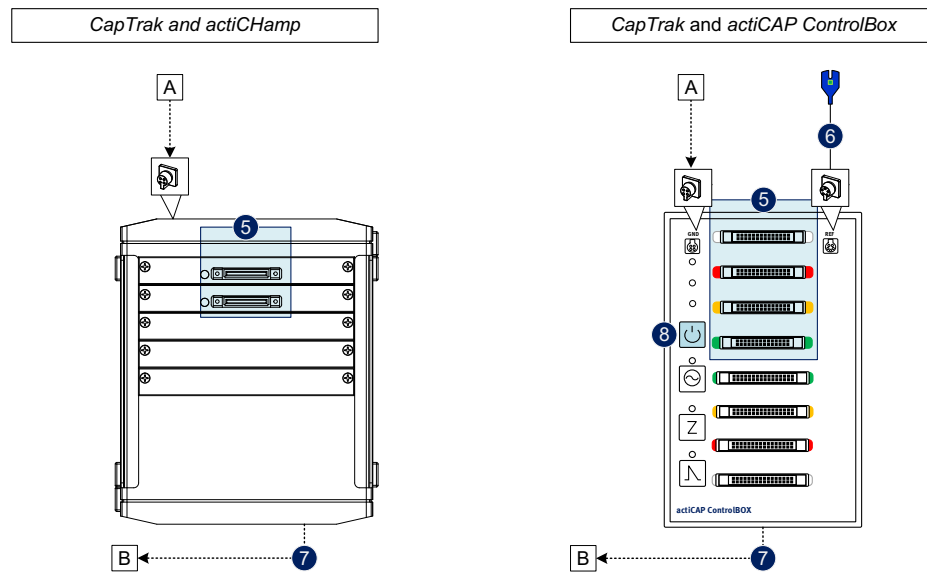


Figure 2-3. Connecting the electrode cap and ground distributor to the electrode input equipment

2.3 Setup with LiveAmp

The following procedure describes the setup of CapTrak with LiveAmp using the LiveAmp actiCAP adapter and active electrodes.



Refer to the LiveAmp Operating Instructions, section 1.2 Optional accessories for LiveAmp, for more information on available adapters.

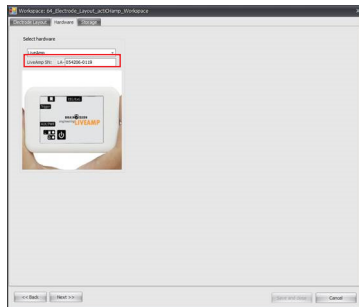
Prepare

- ▶ LiveAmp/s and LiveAmp actiCAP adapter (incl. wireless adapter/s (UBT21)) not supplied
- ▶ landmarks (incl. ground cable and ground distributor (supplied))
- ▶ ground and reference electrodes (not supplied)
- ▶ actiCAP electrode cap (incl. splitter boxes) (not supplied)
- ▶ CapTrak hand-held scanner (supplied)
- ▶ USB license dongle for CapTrak (supplied).

Procedure

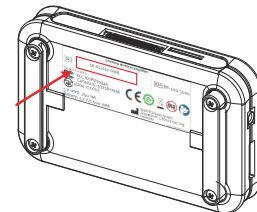
- 1 Connect the hand-held scanner to the computer (rear USB 3.0 port).
- 2 Plug in the USB license dongle.

- 3 Connect the ground electrode to the LiveAmp actiCAP adapter.
- 4 Connect the reference electrode to the LiveAmp actiCAP adapter.
- 5 Connect the electrode cap to the LiveAmp actiCap adapter. Observe the correct order when connecting the splitter boxes.
- 6 Connect the LiveAmp wireless adapter/s (UBT21) to the computer.
- 7 Switch on the computer and run the CapTrak software.
- 8 In workspace wizard then the Hardware tab, select **LiveAmp** from the drop down list.
- 9 Enter the **serial number** of the LiveAmp in CapTrak.



You can find the serial number on the type plate at the bottom of your LiveAmp.

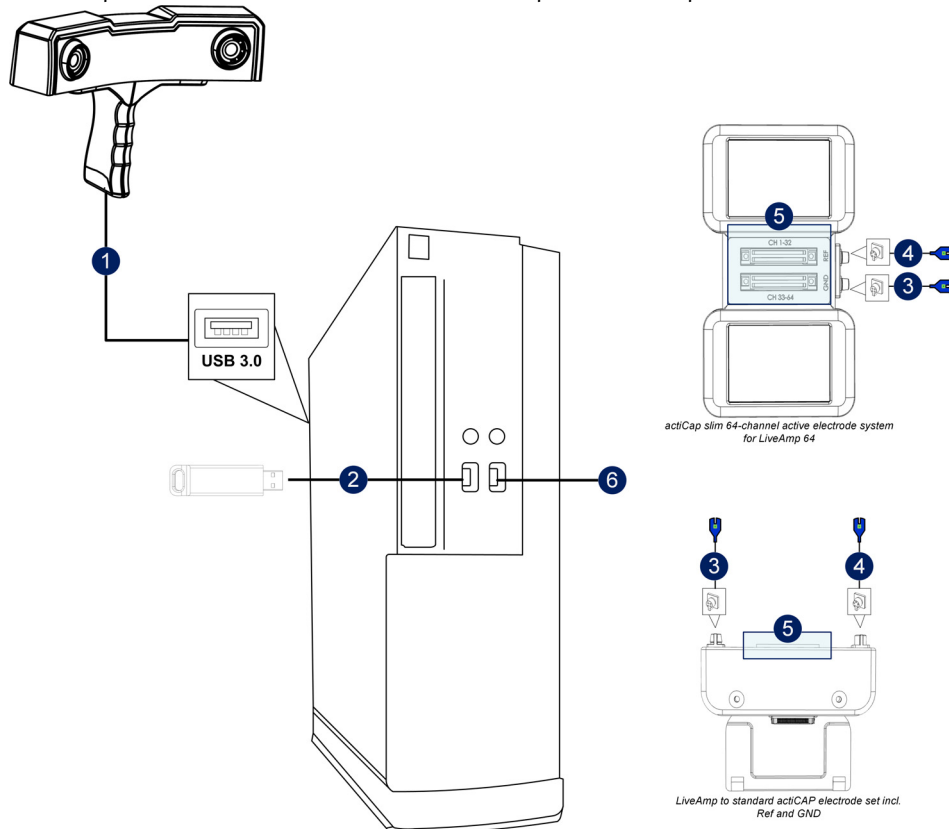
When using LiveAmp 64 enter the serial number of the Master LiveAmp (the LiveAmp connected to CH 1 - 32).



- 10 Turn on LiveAmp/s; CapTrak uses the serial number provided to connect to the corresponding LiveAmp/s. The blue LED on LiveAmp/s will blink to show it is connected to CapTrak.

Results of this task

The setup with active electrodes and the LiveAmp actiCAP adapters is illustrated below.



2.3.1 Attaching the landmarks

The landmarks must be attached to the Nasion, and to the left and right preauricular positions:

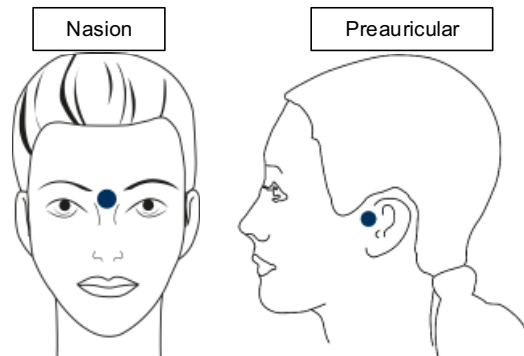


Figure 2-4. Positions for the landmarks

Prepare the following:

- ▶ three landmarks
- ▶ three self-adhesive rings.

Do the following:

- 1 Attach the self-adhesive ring on the subject. Take special care of the position on the subject.
- 2 Remove the protective film of the self-adhesive ring.
- 3 Attach the landmark onto the self-adhesive ring.

Alternatively, you can first attach the self-adhesive ring to the landmark, and then attach the landmark to the subject.

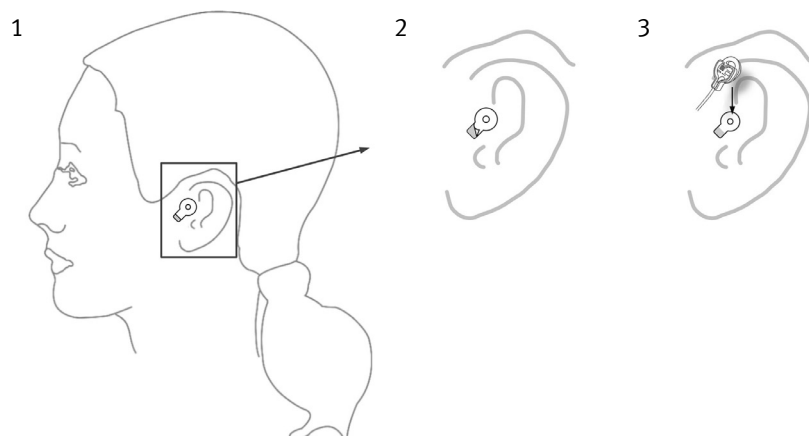


Figure 2-5. Attaching the landmarks

2.4 Normal use

The normal use is the scanning of the EEG electrodes and landmarks. Observe the following hints for scanning:

- ▶ You must be able to move around the subject in order to scan all electrodes. Make sure, that there is sufficient place to move.
- ▶ The cable of the hand-held scanner is approximately four meters long. Position the test subject well within the radius of the cable.
- ▶ You can move the scanner during scanning. Gentle, steady movements can improve the scanning process.

2.5 After use

NEW

In between measurements, we recommend using the foam inlet of the CapTrak case to put the hand-held scanner in its designated place, or to lay it gently(!) with the lenses on the anti-slip mat. This prevents the scanner from unintentional movements.

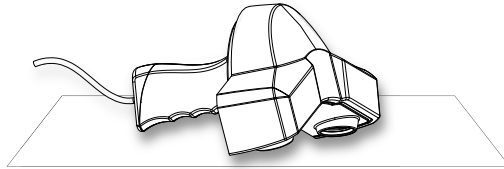


Figure 2-6. Storing the hand-held scanner

**Note**

For storing the hand-held scanner, put it into its designated place of the CapTrak case. Observe the ambient conditions for storing the hand-held scanner.



Chapter 3

Using the software

3.1 Installing the software

Upon delivery *CapTrak* is installed on the computer.



Note

When delivered the power plan on the computer is set to High performance, modifying this may cause issues for *CapTrak*.

If you are installing *CapTrak* on another computer or would like to verify the power plan settings open the **Control Panel**, select **Power Options** then **Create a power plan**. Ensure **High performance** is selected as seen in [Figure 3-2](#).

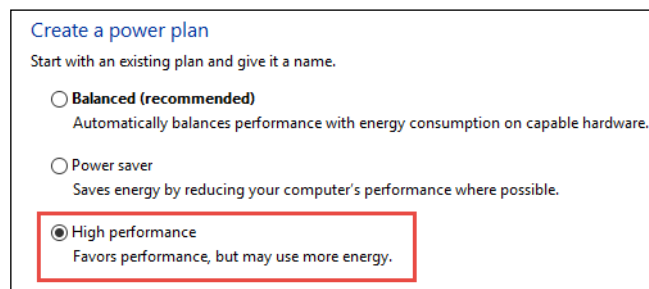


Figure 3-1. High performance selected in power plan

You can reinstall *CapTrak* on the computer, if necessary. You can also install *CapTrak* on a different computer. Please observe the system requirements in [Appendix A](#).

Prepare the following items:

- ▶ *Application Suite USB* (supplied)
- ▶ hand-held scanner (supplied)
- ▶ license dongle (supplied)
- ▶ USB stick with the calibration file (supplied).

Do the following:

- 1 Open the *Application Suite USB*.
- 2 Click on **Install BrainVision CapTrak > Install BrainVision CapTrak**.
- 3 All necessary components will be installed automatically.

- 4 Reboot your computer when prompted.

After installing

After installing you must load the scanner calibration file into *CapTrak*. Do the following:

- 1 Connect the hand-held scanner to the rear USB 3.0 port of the computer.
- 2 Connect the license dongle and USB stick with the calibration file to the computer.
- 3 Start *CapTrak*.
- 4 To load the calibration file from the USB stick:
 - ▷ Click on **Calibration > Load Calibration from File**.
 - ▷ Select the calibration file from the USB stick and follow the on-screen messages.

3.2 Normal use

Log into the user account that is called **CapTrak**.

Starting the software

Do the following:

- 1 Plug in the USB dongle (supplied).
- 2 Prepare the hardware as described in [Chapter 2](#).
- 3 Start the software via the desktop shortcut .

The following areas are on the user interface:

- 1 **Menu** bar
- 2 **Scanning** bar
- 3 **Scanner distance** window
- 4 Scanner images
- 5 **Scan Settings** window
- 6 **Notification** window
- 7 **Log** window
- 8 **Status** bar.

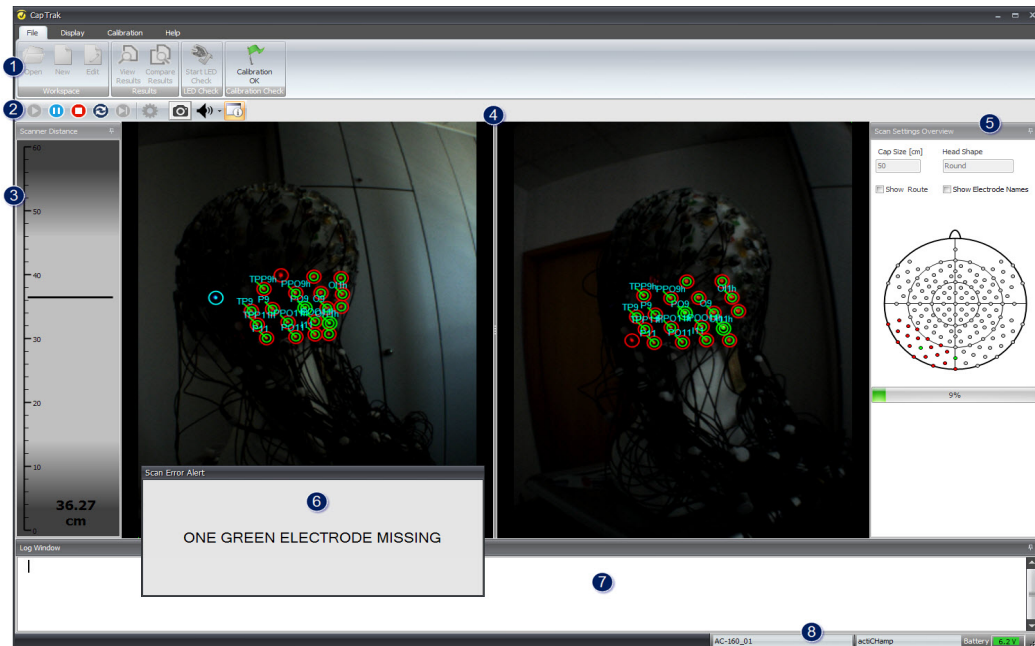


Figure 3-2. CapTrak user interface at a glance



Notes for first-time users

Always plug in the hand-held scanner and license dongle before you start the software.

Basic principles of operation

CapTrak needs some basic information for scanning. The information consists of the electrode layout and used electrode input equipment. These details are stored in **workspaces**. Thus, before you can start scanning, you must configure a workspace. Workspaces usually remain unchanged throughout your project.

The following information is stored in a workspace:

- ▶ electrode layout (provided by the cap manufacturer)
- ▶ electrode input equipment
- ▶ file name and file directory settings.



Note

You can find pre-configured standard workspaces on the *Application Suite USB*.



For detailed information, please refer to the help topics in the software.

3.3 Notes for system administrators



Notes on the administrator account

The computer has a default administrator account:

- ▶ **Log in:** Admin
- ▶ **Password:** admin



Notes on the dongles

The dongle with the **UC** serial number contains a full license for *CapTrak*. The dongle with the **URA** serial number contains a 2 months trial license for the *Application Suite USB*, for example, *Recorder*, *Analyzer*, *RecView*.

For more information please visit:

<http://www.brainproducts.com/productdetails.php?id=59>





Chapter 4

Maintenance and cleaning

4.1 Calibration check of the scanner

The hand-held scanner was calibrated before delivery. The calibration check ensures that the scanner still has the factory calibration.

4.1.1 *Prepare the environment for a calibration check*

The calibration check can be strongly influenced by environmental conditions and due to this a failed calibration check does not necessarily mean that the system is out of calibration. The following may influence the results of a calibration check:

- ▶ shadows on the calibration checkerboard
- ▶ strong light shining on the calibration checkerboard
- ▶ ambient light (either too dark or too bright)
- ▶ movement in the hand-held scanner while the calibration check is in progress

For example, if the environmental conditions are not optimal, running a calibration check ten times may result in only one successful calibration check. Nevertheless, one successful calibration check does confirm that the scanner is correctly calibrated.

4.1.2 *Perform a calibration check*

In order to ensure precise results, regularly check the calibration of the hand-held scanner.

Prepare the following:

- ▶ hand-held scanner
- ▶ license dongle
- ▶ calibration checkerboard
- ▶ anti-slip mat.

Do the following:

- 1 Connect the hand-held scanner and license dongle to the computer.

- 2 Set the calibration checkerboard in portrait orientation on the anti-slip mat.



Note

Ensure that the ambient light is evenly distributed across the board to reduce shadows or dark areas.

- 3 Open *CapTrak* and go to **Calibration > Start Check**.
- 4 Point the hand-held scanner at the center of the checkerboard at a height of approximately 30cm and ensure that it is held steady during the calibration.

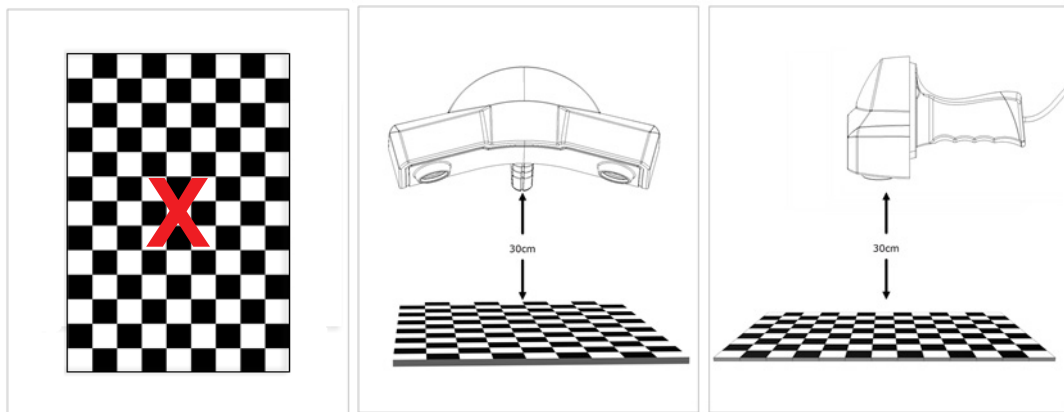


Figure 4-1. Calibration check

- 5 Compare the displayed patterns with the example patterns in the **Calibration** window. On the right hand side of the screen, two squares indicate whether the pattern is detected and accepted.
- 6 The outcome of the calibration check is shown in a message box.
 - ▷ Successful calibration check - a tone will sound and the message will state that the calibration check has been successful.
 - ▷ Unsuccessful calibration check - the message will state that the calibration check has been unsuccessful and to try again.

After three attempts without a successful calibration check please modify the **Gain** and **Shutter time** settings to give a clearer image on screen. Click  to open [Figure 4-2](#) Gain and Shutter time settings.

After several further attempts without success, please contact your local dealer.

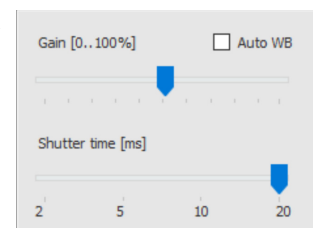


Figure 4-2. Gain and Shutter time settings



For further information, please refer to the help topics in the software (.

**Note**

A USB stick with the original calibration file is in the scope of delivery. The USB stick is labeled with the serial number of the hand-held scanner. Use the USB stick, if you are prompted to load the original calibration.

4.2 Cleaning

Clean the components of the *CapTrak* family with a slightly damp cloth.

Be very careful when cleaning the lenses. Use a soft microfiber cloth and do not apply pressure onto the lenses.

**Notice**

Do not allow water to enter the hand-held scanner.

Do not use aggressive, scratching or acidic cleaning agents.

4.2.1 Using a disinfectant

For disinfecting the surfaces of products (amplifiers, batteries, adapter boxes, etc.), we recommend wiping them down with a tissue drenched in a 70% concentration of isopropyl alcohol (commonly called isopropanol or 2-propanol) for at least 30 seconds.

If you are unable to purchase the products mentioned above in your region, a disinfecting product with similar proportions and active ingredients would be recommended.

4.3 Disposal

NEW

For disposal of the hand-held scanner and its accessories please observe the applicable legislation. In the EU and EFTA, the Directive on Waste Electrical and Electronic Equipment (WEEE) 2012/19/EU as well as the Directive on Batteries and Accumulators and Waste Batteries and Accumulators (Battery Directive) 2006/66/EC apply. In Germany, for example, the 'ElektroG' and the 'BatteryG' additionally govern electrical and electronic equipment.

Battery type: Lithium metal coin battery

Batteries are not accessible and must be disposed with the dongle.

Do not dispose of your devices, accessories and cables with the household waste. Dispose of them in accordance with national regulations or return the dongle to the manufacturer.

On request, Brain Products will take back your old devices, provided that they are original Brain Products parts.

REACH SVHC notification

In accordance with Regulation (EC) No. 1907/2006 of the European Parliament and of the Council of 18 December 2006 concerning the Registration, Evaluation, Authorization and Restriction of Chemicals (REACH) as defined in article 57 Brain Products herewith notifies that the coin battery in the dongles may contain a concentration of a substance of very high concern (SVHC) above 0.1 % weight.

Substance	CAS No.
1, 2-dimethoxyethane; ethylene glycol dimethyl ether (EGDME)	110-71-4

Data protection: We would like to point out to all end users of the products that you are responsible for deleting personal data on the products to be disposed of.





Appendix A Technical data

Ambient conditions

Operation	Temperature range: 10 °C to +40 °C (50 °F to 104 °F) Relative humidity: 10 % to 85 %, non-condensing Atmospheric pressure range: 700 hPa to 1,050 hPa
Transport and storage	Temperature range: 0 °C to +60 °C (32 °F to 140 °F) Relative humidity: 10 % to 85 %, non-condensing Atmospheric pressure range: 700 hPa to 1,050 hPa

Scanner data

Connection	USB 3.0
Supply voltage and power	max. 4 W (at 5 VDC)
Dimensions	28 x 18.5 x 13 cm (L/H/D)
Weight	approx. 840 g

Minimum system requirements

Operating system	Windows® 10
RAM	4 GB
Available hard-disc space	70 MB
CPU	3.4 GHz
USB	USB 3.0 or higher



Appendix B Reading results correctly

The resulting electrode positions of your scan can be used to construct a real model of the head.

In general, a scan results in the precise **Cartesian** and **spherical** coordinates for each electrode.



Note

NEW

CapTrak detects the electrodes' LED positions. The distance between scalp surface and LED is not considered in the final result. The distance between scalp surface and LED mainly depends on thickness of hair and the LED position within the electrode (for actiCAP slim electrodes ~3.6 mm and additional 2 mm when actiCAP snap holders are used)

Cartesian coordinates

The underlying 3-axis coordinate system is constructed through the landmarks. They determine the position of the coordinate system as follows:

- ▶ X-axis: Line through LPA (left preauricular) and RPA (right preauricular), where LPA is located on the negative axis.
 - ▶ Y-axis: Orthogonal line through the Nasion landmarks onto the x-axis.
 - ▶ Z-axis: Orthogonal line on xy-plane through Origin (O)
- The intersection of the x- and y-axes is the Origin(O) (from the xy-plane)

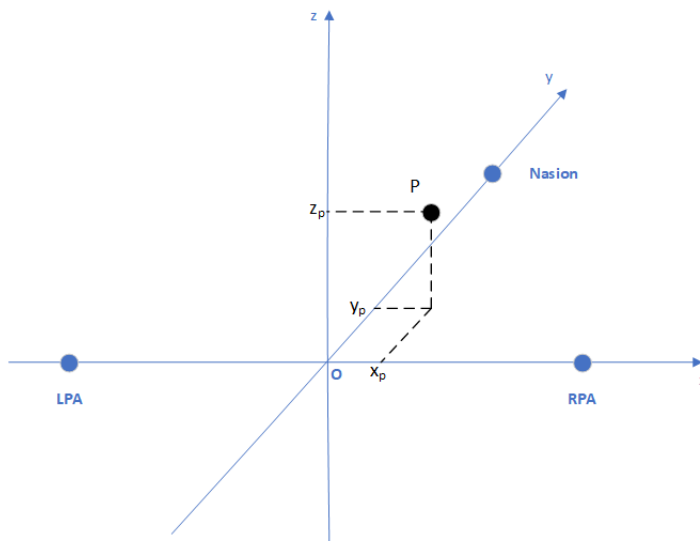


Figure B-1. Cartesian coordinate system

In the results list, the Cartesian coordinates are shown by the x, y, z values.

Result	Cap Position					
Electrode	X [mm]	Y [mm]	Z [mm]	Phi [°]	Theta [°]	Radius [mm]
AF3	-41.67	115.75	48.5	-70.2	-68.48	132.23
AF4	37.49	119	51.97	72...	67.39	135.16
AF7	-62.37	100.58	16.88	-58.2	-81.88	119.54
AF8	61.27	99.2	26.64	58.3	77.13	119.6
AFF10h	65.29	78.68	0.12	50...	89.93	102.24
AFF11h	-68.32	63.33	-33.83	-4...	-109.96	99.11
AFF12h	65.58	64.23	-34.77	44.4	110.74	98.16
AFF1h	-25.18	118.79	77.44	-7...	-57.47	144.02
AFF2h	18.62	120.2	78.65	81...	57.11	144.84
AFF5h	-59.28	101.89	44.77	-5...	-69.2	126.09
AFF6h	56.4	103.95	50.23	61...	66.99	128.49
AFF9h	-69.79	79.02	-3.69	-4...	-92.01	105.5
AFp1	-20.69	126.05	40.04	-8...	-72.6	133.86
AFp2	16	127.22	41.2	82...	72.19	134.68
AFz	-5.5	126.53	61.79	-8...	-63.99	140.92
C1	-44.47	38.03	142.21	-4...	-22.36	153.78
C2	33.24	40.63	146.18	50...	19.75	155.32
C3	-75.73	28.67	108.81	-2...	-36.66	135.63
C4	71.3	32.28	117.58	24...	33.65	141.25
C5	-80.43	50.46	58.92	-32.1	-58.18	111.74
C6	84.12	22.23	79.05	14.8	47.74	117.55
CCP1h	-27.1	20.76	154.42	-3...	-12.47	158.15
CCP2h	15.49	22.45	156.04	55.4	9.91	158.41
CCP3h	-64.93	14.07	130.56	-1...	-26.97	146.49
CCP4h	57.12	17.76	138.63	17...	23.34	150.99

Figure B-2. Cartesian coordinates

When you scroll through the results you will notice that some of the values are '0' (zero). This concerns the preauricular landmarks LPA, RPA and Nasion landmark.

Spherical coordinates

The spherical coordinates are the positions on the surface of the scalp. They are determined by using the underlying spherical coordinates as illustrated below.

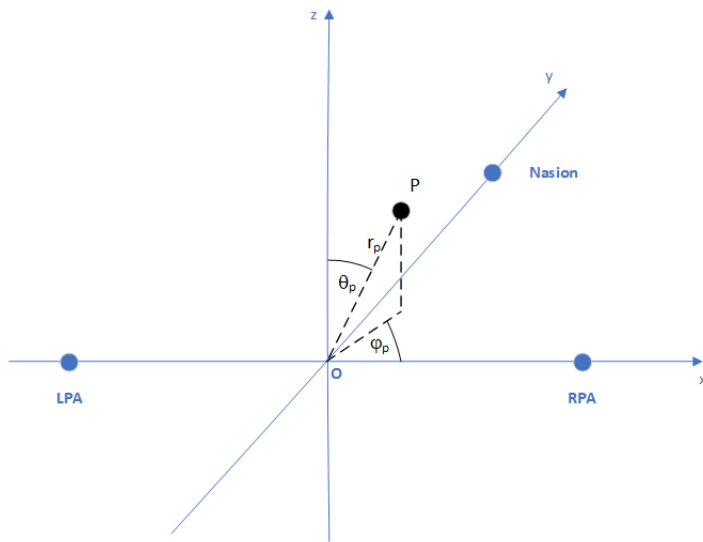


Figure B-3. Spherical coordinate system

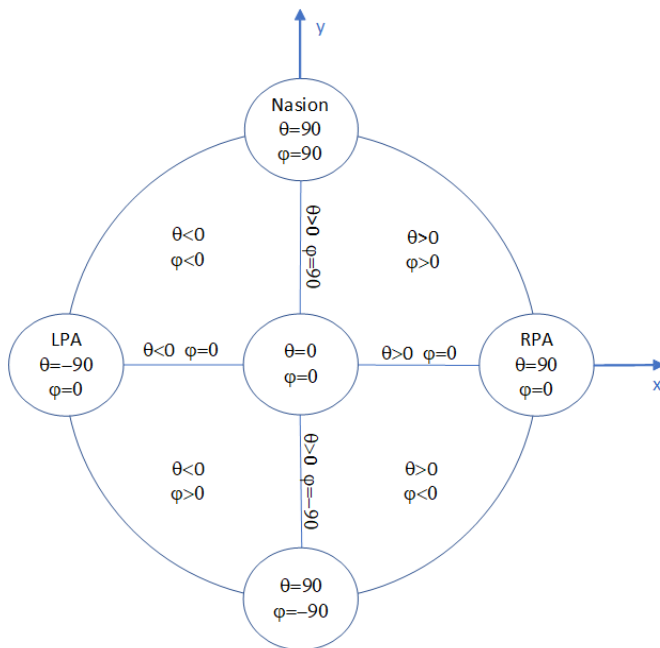


Figure B-4. Sign convention for angles

In the results list, the spherical coordinates are shown by the phi, theta and radius values.

Result	Cap Position					
Electrode	X [mm]	Y [mm]	Z [mm]	Phi [°]	Theta [°]	Radius [mm]
AF3	-41.67	115.75	48.5	-70.2	-68.48	132.23
AF4	37.49	119	51.97	72...	67.39	135.16
AF7	-62.37	100.58	16.88	-58.2	-81.88	119.54
AF8	61.27	99.2	26.64	58.3	77.13	119.6
AFF10h	65.29	78.68	0.12	50...	89.93	102.24
AFF11h	-68.32	63.33	-33.83	-4...	-109.96	99.11
AFF12h	65.58	64.23	-34.77	44.4	110.74	98.16
AFF1h	-25.18	118.79	77.44	-7...	-57.47	144.02
AFF2h	18.62	120.2	78.65	81...	57.11	144.84
AFF5h	-59.28	101.89	44.77	-5...	-69.2	126.09
AFF6h	56.4	103.95	50.23	61...	66.99	128.49
AFF9h	-69.79	79.02	-3.69	-4...	-92.01	105.5
AFp1	-20.69	126.05	40.04	-8...	-72.6	133.86
AFp2	16	127.22	41.2	82...	72.19	134.68
AFz	-5.5	126.53	61.79	-8...	-63.99	140.92
C1	-44.47	38.03	142.21	-4...	-22.36	153.78
C2	33.24	40.63	146.18	50...	19.75	155.32
C3	-75.73	28.67	108.81	-2...	-36.66	135.63
C4	71.3	32.28	117.58	24...	33.65	141.25
C5	-80.43	50.46	58.92	-32.1	-58.18	111.74
C6	84.12	22.23	79.05	14.8	47.74	117.55
CCP1h	-27.1	20.76	154.42	-3...	-12.47	158.15
CCP2h	15.49	22.45	156.04	55.4	9.91	158.41
CCP3h	-64.93	14.07	130.56	-1...	-26.97	146.49
CCP4h	57.12	17.76	138.63	17...	23.34	150.99

Figure B-5. Spherical coordinates

Electrode positions in single results

Results contain the **electrode positions** and information about the cap position.

Electrode positions

The electrode positions consist of the **Cartesian** (x, y, z) and **spherical** coordinates (phi, theta, radius) for each electrode.

Result	Cartesian			Spherical		
	Cap	Position				
Electrode	X [mm]	Y [mm]	Z [mm]	Phi [°]	Theta [°]	Radius [mm]
FT10	79.16	27.78	-6.06	19.34	94.13	84.11
TP12	71.85	-39.57	-19.07	-28...	103.09	84.22
FT12	69.43	20.37	-45.17	16.35	121.98	85.3
RPA	85.53	0	0	0	90	85.53
LPA	-86.63	0	0	0	-90	86.63
TP11	-72.37	-39.91	-27.77	28.88	-108.57	87.19
TPP11h	-70.43	-51.32	-4.45	36.08	-92.92	87.26
TP10	75.51	-40.89	17.91	-28...	78.22	87.72
FT9	-82.7	27.84	-10.08	-18.6	-96.59	87.84
TPP12h	71.42	-51.48	3.12	-35...	87.97	88.1
FFT12h	73.37	40.35	-31.09	28.81	110.37	89.32
FTT9h	-86.37	22.12	9.87	-14...	-83.68	89.7
FTT10h	85.28	22.54	17.58	14.8	78.73	89.94
TP9	-80.07	-42.08	6.76	27.72	-85.73	90.71
FFT11h	-77.09	37.82	-31.81	-26...	-110.33	91.57
P9	-71.24	-56.99	18.25	38.66	-78.69	93.04
FT11	-76.35	19.52	-49.58	-14...	-122.18	93.1
P11	-64.84	-65.98	-13.5	45.5	-98.3	93.48
F12	63.45	46.62	-50.8	36.31	122.83	93.7
FFT10h	78.38	51.23	12.36	33.17	82.48	94.45
F10	73.97	58.52	-11.14	38.35	96.74	94.97
FFT9h	-81.28	48.64	6.99	-30.9	-85.78	94.98
P10	71.54	-56.18	27.79	-38...	73.01	95.11
PPO11h	-58.39	-74.56	11.85	51.94	-82.87	95.44
T7	-88.79	11.35	34.2	-7.28	-69.09	95.83

Figure B-6. Results

Cap position

The cap position indicates the deviation of the electrodes from the sagittal midline.

Result

Cap Position

Electrode deviation from sagittal midline

Electrodes	Distance [mm]
Nasion	0
AFz	-6.52
Fz	-7.29
FCz	-9.66
Cz	-10.02
CPz	-13.18
Pz	-19.24
POz	-13.3
Oz	-14.15

Electrode distances on sagittal midline

Electrode Pairs	Measured Dist. [mm]	Optimal Dist. [mm]
Nasion -> AFz	62.11	85.87
AFz -> Fz	40.78	42.45
Fz -> FCz	42.92	42.45
FCz -> Cz	43.31	44.36
Cz -> CPz	44.1	42.45
CPz -> Pz	44.94	44.36

Figure B-7. Cap position



Cap position information

This information is not displayed, when you compare results.

Electrode positions in compared results

You can compare two scans, for example, to find out if the cap has moved.

At the bottom of the **Compare** tab, there are the three values **Distance**, **Mean distance** and **Median distance**.

Result	Compared result	Comparison		
Electrode	Distance [mm]	Diff. X [mm]	Diff. Y [mm]	Diff. Z [mm]
FT10	1.218	1.205	0.057	0.168
FT7	1.391	0.503	0.681	1.103
FT8	1.293	1.157	0.568	0.108
FT9	1.518	0.604	0.868	1.09
FTT10h	1.546	1.434	0.539	0.21
FTT7h	1.77	1.039	0.818	1.177
FTT8h	2.161	2.057	0.574	0.328
FTT9h	1.464	0.595	0.759	1.102
Fz	1.312	0.996	0.714	0.468
GND	1.035	0.374	0.877	0.403
Iz	8.098	1.252	2.156	7.705
Nasion	0.952	0	0.952	0
O1	3.241	1.605	0.327	2.796
O10	3.98	1.132	0.389	3.796
O2	4.111	1.753	1.08	3.558
O9	5.444	2.364	1.14	4.77
OI1h	6.678	4.637	0.223	4.8
OI2h	5.401	1.134	2.987	4.354
Oz	3.445	0.911	0.46	3.29
P1	2.938	1.785	2.019	1.172
P10	2.052	1.881	0.291	0.768
P2	2.518	1.979	1.449	0.569

Distances ≥ 5 mm: 4
Mean distance: 2.153 mm.
Median distance: 2.089 mm.

Figure B-8. Comparison

- Distances ≥ 5 mm shows the number of electrodes with a distance of 5 mm or more to their compared position.
- Mean distance is the average value of all distances.
- Median distance is exactly the value of the distance that is in the middle of all distances when sorted in ascending order (this means 50 % of the distance values are lower than the median distance and 50 % of the distance values are higher than the median distance). The median distance is more stable against outliers.

Cap position

Shows the accuracy of the cap position with the help of the sagittal midline.

Electrode deviation from the sagittal midline

The deviation of the electrodes from the sagittal midline is visualized by the amber line.

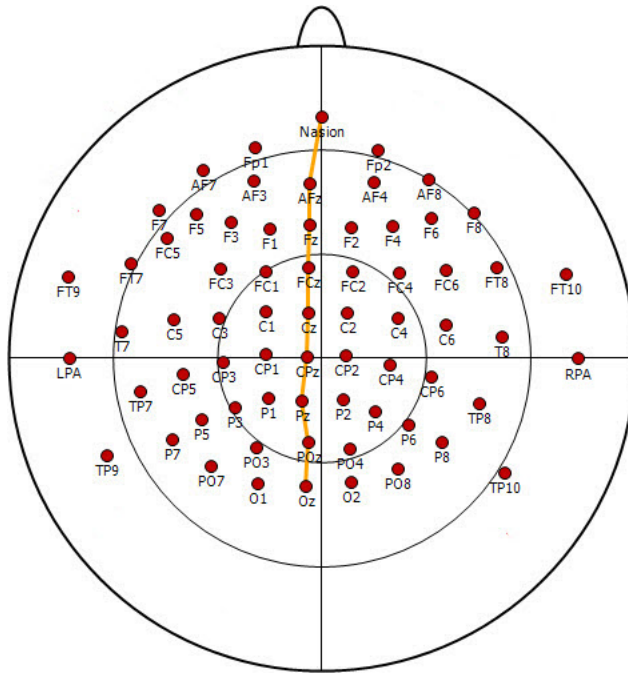


Figure B-9. Sagittal Line

The deviation is presented in a list, where negative values indicate a deviation to the left of the midline and positive values to the right of the midline (seen from above).

Electrode deviation from sagittal midline	
Electrodes	Distance [mm]
Nasion	0
AFz	-6.52
Fz	-7.29
FCz	-9.66
Cz	-10.02
CPz	-13.18
Pz	-19.24
POz	-13.3
Oz	-14.15

Figure B-10. Electrode deviation from sagittal midline

Electrode distances on the sagittal midline

The electrode distances on the sagittal midline is presented in pairs with the measured and optimal distance. The values for the optimal distance are based on the cap layout.

The calculation uses the electrodes on the sagittal midline from the Nasion to the Pz position.

Electrode distances on sagittal midline		
Electrode Pairs	Measured Dist. [mm]	Optimal Dist. [mm]
Nasion -> AFz	62.11	85.87
AFz -> Fz	40.78	42.45
Fz -> FCz	42.92	42.45
FCz -> Cz	43.31	44.36
Cz -> CPz	44.1	42.45
CPz -> Pz	44.94	44.36

Figure B-11. Electrode distances from sagittal midline

The first value in the list, is the distance between the Nasion (landmark) and the first electrode on the midline, for example AFz. The last value always ends with the Pz position.

Large deviations might be an indication of a high tension on the cap material, that is, the cap is probably too small. If this is the case, measure the head circumference of the subject again and re-assure that you have used the correct cap size.

Improper electrode positions

CapTrak can recognize wrongly placed electrodes.

We distinguish between two **switched** electrodes within a pattern, and more than two switched electrodes in different patterns. The latter one is designated as **misplaced** electrodes.



Improper electrode positions

Always confirm the correct positions of the electrodes in the model before your measurement. If you find improperly placed electrodes, correct them on the cap.

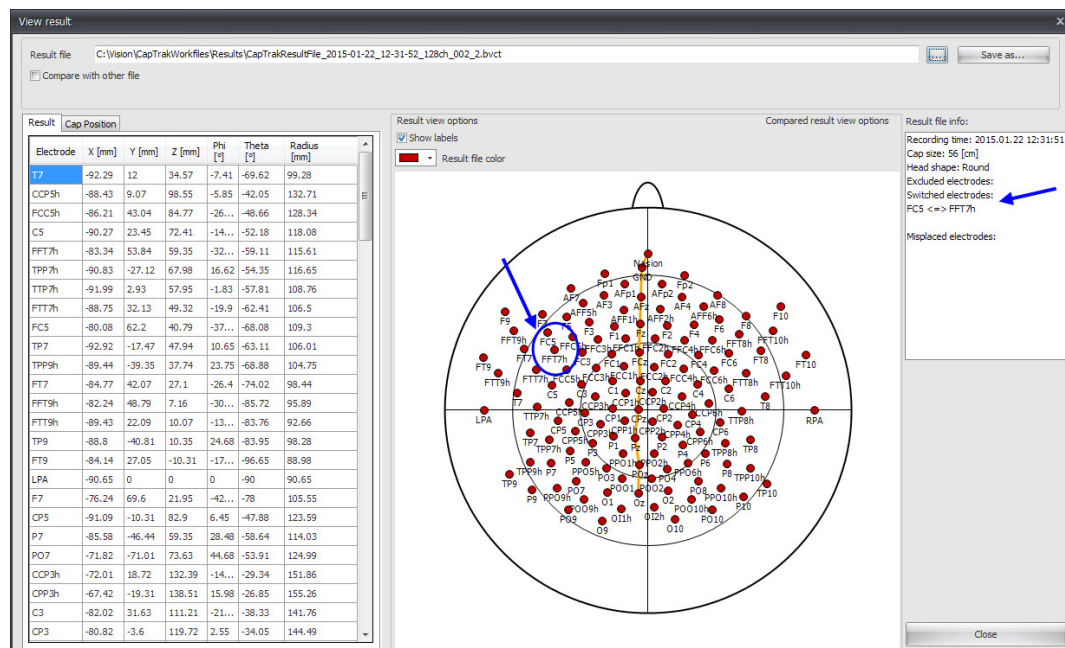
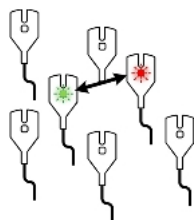


Figure B-12. View result

Switched electrodes



Switched electrodes in a pattern are displayed.

As soon as the software detects switched electrodes, the scanning process stops and the two electrodes blink.

Additionally, a message indicates the switched electrodes:

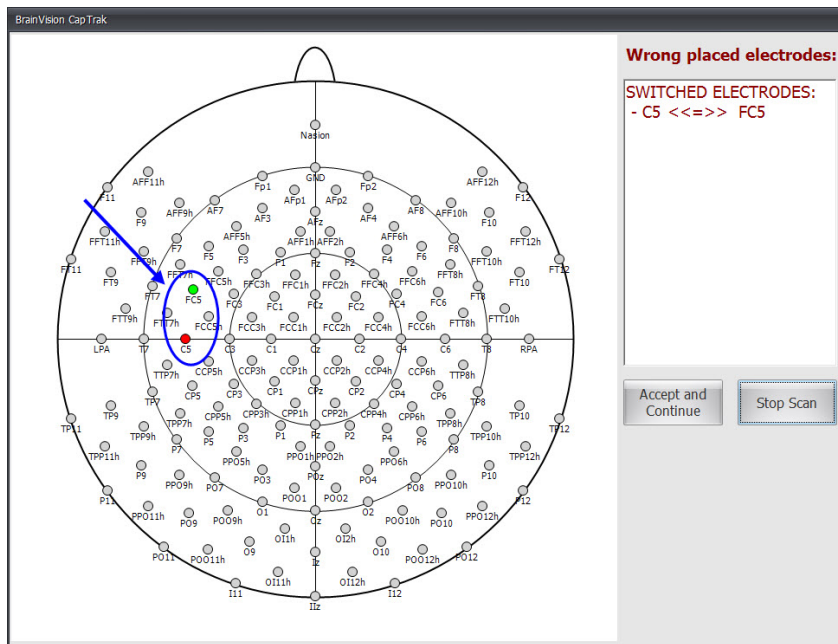
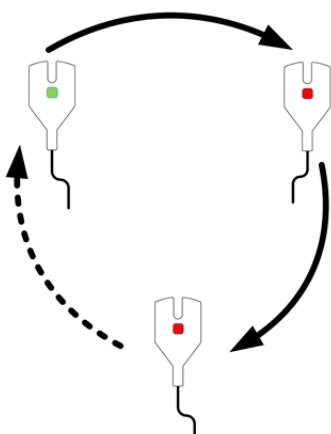


Figure B-13. Switched electrodes message

If you have intentionally switched the electrodes, then click on Accept and Continue.

Otherwise correct the switched electrodes and then click on Stop Scan.

Misplaced electrodes



If the software detects electrodes that are not on their expected position, they will be identified as 'misplaced electrodes'.

Misplaced electrodes are three or more electrodes that are not on their expected place.

During scanning, you can detect potentially misplaced electrodes by comparing the actual pattern with the target pattern:



Figure B-14. Compare actual pattern and target pattern

If the software detects misplaced electrodes, the electrodes will be indicated at the end of the scan in the View result window.

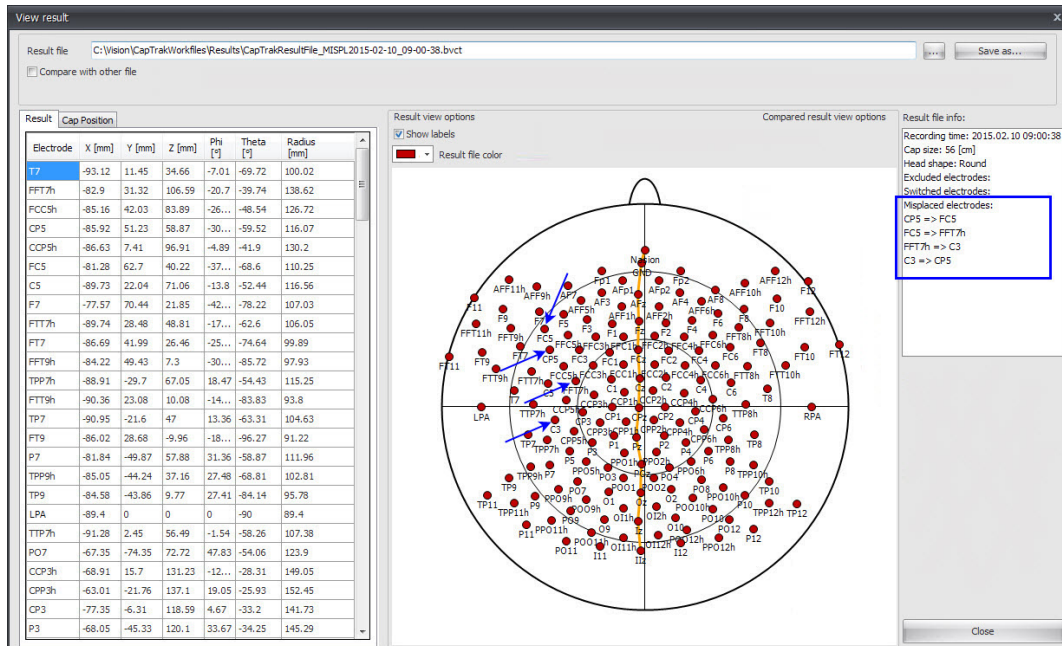


Figure B-15. View result window

If you have not intentionally changed their positions, correct the error and perform a second scan.



Check the result file info

Always check the result file info, in order to avoid improperly placed electrodes.